

B. Sc. (Hons.) III SEMESTER MID-TERM EXAMINATION, 2022-23
STATISTICS

Paper No. STB-301: STATISTICAL INFERENCE

Time: One hour

Full Marks: 20

Note: Answer any two questions. All questions carry equal marks.

1. (a) What do you mean by sampling distribution? Let X be a random variable with expectation $E(X) = \mu$ and variance $V(X) = \sigma^2$. Let $x_1, x_2, \dots, x_n$ be a random sample of size n from the distribution of X . Then show that:

$$E(\bar{x}) = \mu$$

$$V(\bar{x}) = \frac{\sigma^2}{n}$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n X_i$.

- (b) Define point and interval estimates. Let $X_1, X_2, \dots, X_n$ be n independent binomial variables with parameters $(m_1, p), (m_2, p), \dots, (m_n, p)$ respectively. Derive the distribution of $\sum_{i=1}^n X_i$.
2. (a) Define simple and composite hypotheses with examples. Also define level of significance and power function. Let p be the probability that a coin will fall head in a single toss. In order to test $H_0 : p = 1/2$ against $H_1 : p = 2/3$, the coin is tossed thrice and H_0 is rejected if three heads appear. Find the values of α , β , and the power function of the test.
- (b) Define Chi-square statistic and discuss its properties. Describe the chi-square test of independence of attributes in a contingency table.
3. (a) Define Student's t -statistic and discuss its properties. A random sample of size n is drawn from a bivariate normal population. Describe a suitable test for testing the null hypothesis $H_0 : \rho = 0$ against the alternative hypothesis $H_1 : \rho > 0$, where ρ is the population correlation coefficient.
- (b) Show that the standard Cauchy distribution is a special case of Student's t -distribution. Two independent random samples of sizes n_1 and n_2 are drawn from normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively. Describe a suitable test for testing the hypothesis $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$ when σ_1^2 and σ_2^2 are equal and unknown.
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External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>